

Experiment No: 03

Working with Amazon DynamoDB

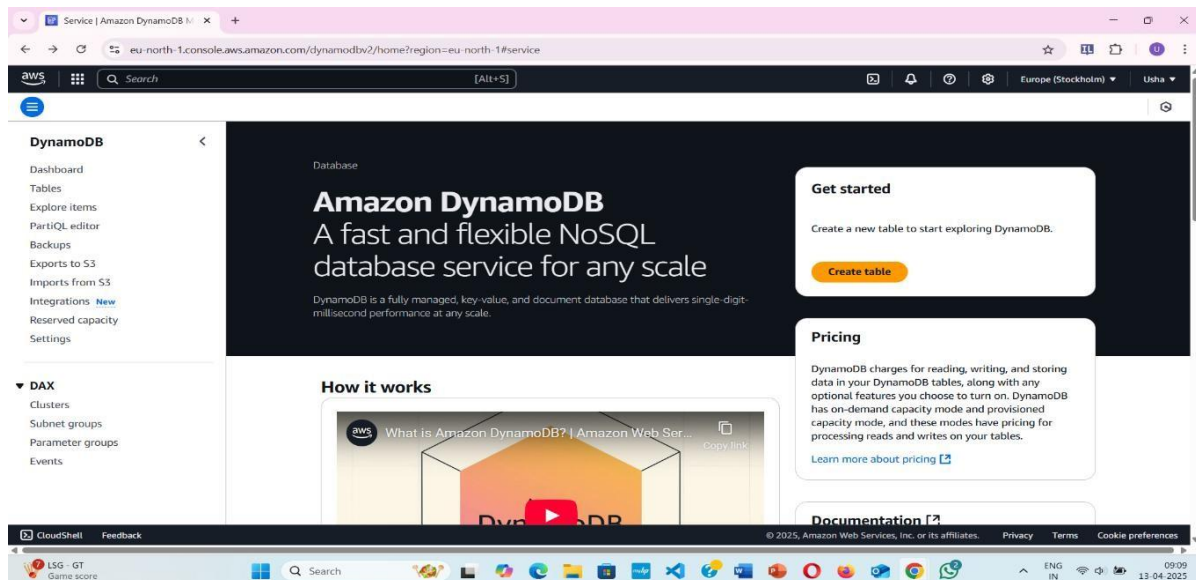
Apparatus Required: AWS Account, AWS Dynamo DB

Pre-Requisite: Aws basics, DBMS basics

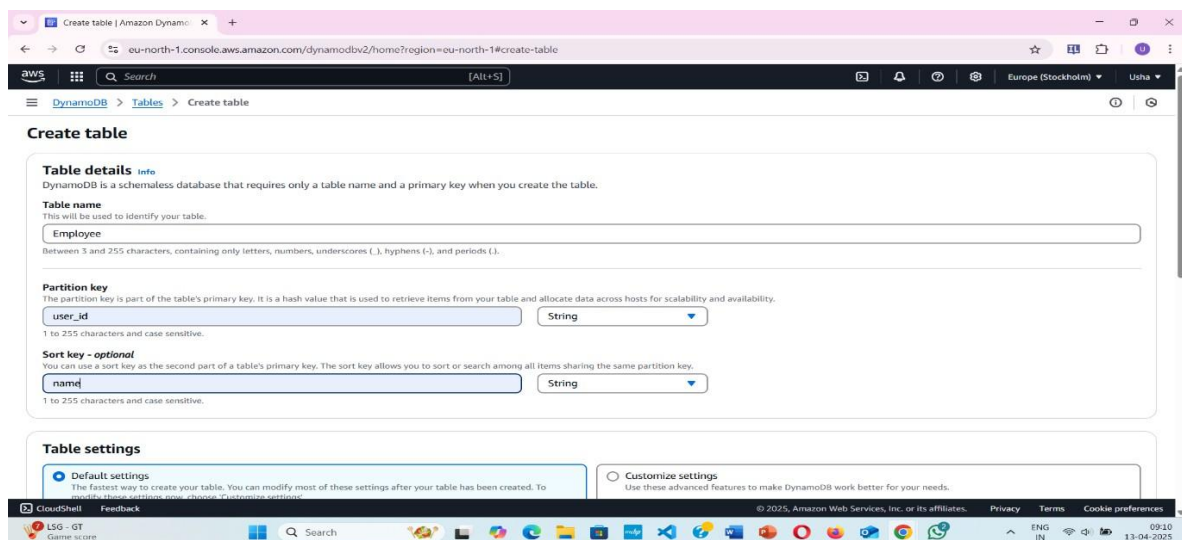
Introduction: AWS DynamoDB is a fast and flexible NoSQL database service for all applications that need consistent, single-digit-millisecond latency at any scale. Its flexible data model and reliable performance make DynamoDB a great fit for mobile, web, gaming, advertising technology, Internet of Things, and other applications.

Procedure

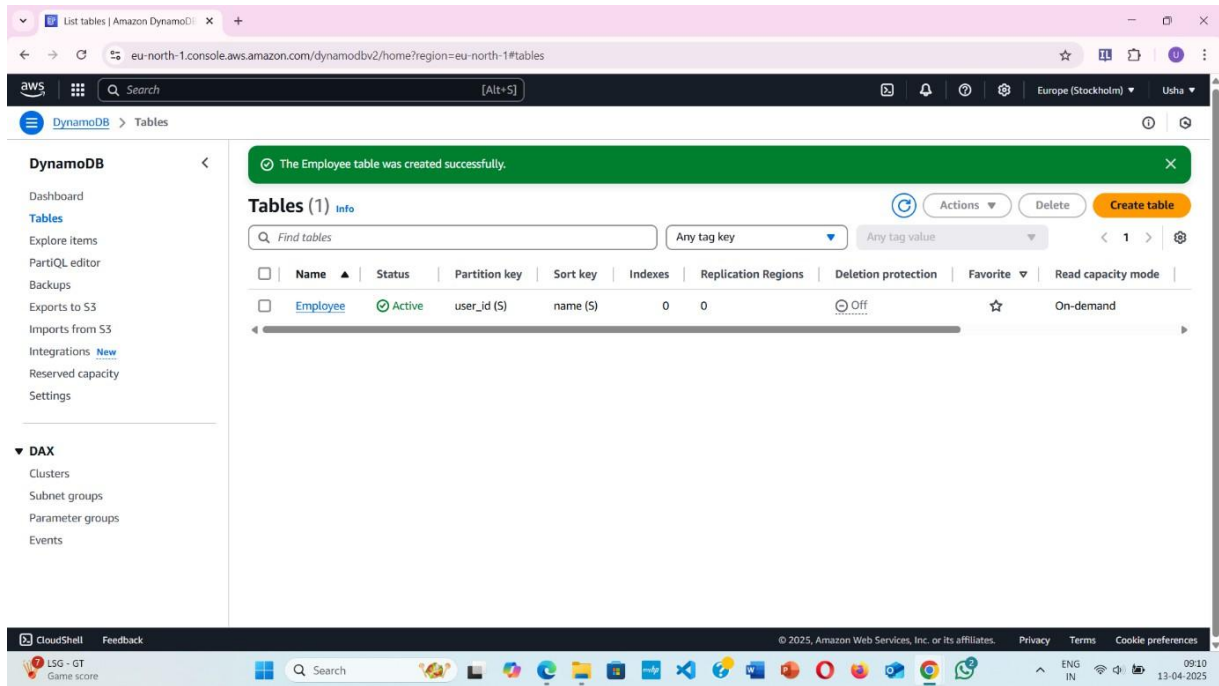
Navigate to Amazon DynamoDB→click on create table



Give table name, Partition Key (is a primary key)



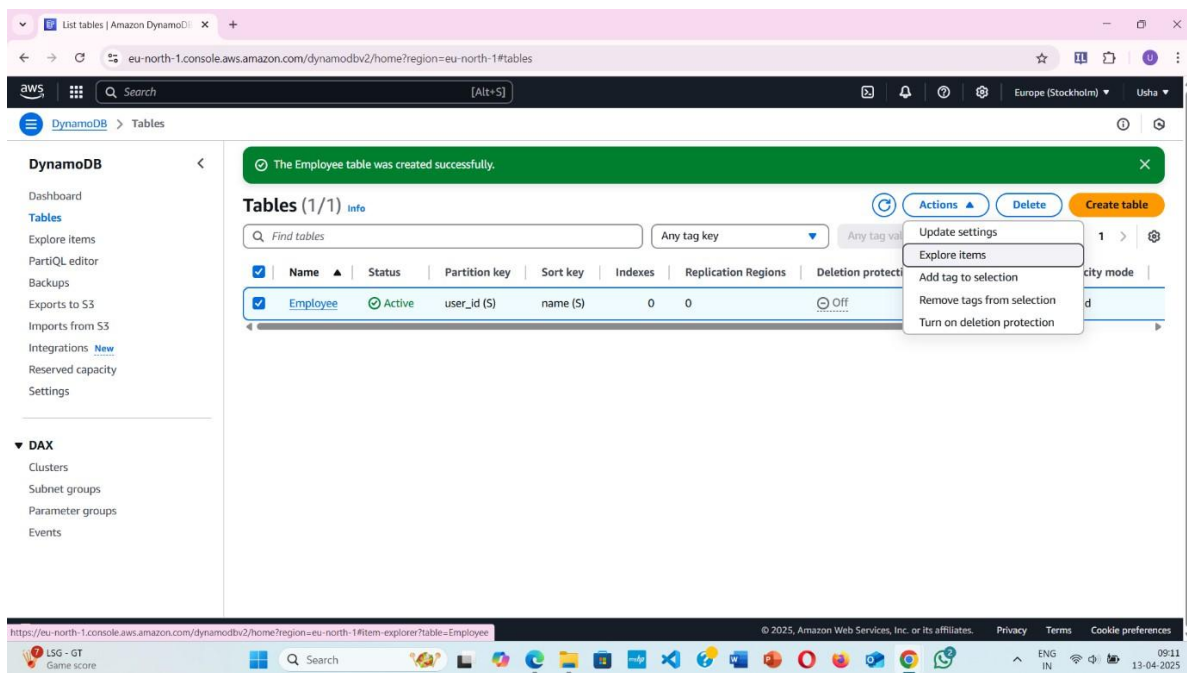
Successfully created the table



The screenshot shows the AWS Management Console for Amazon DynamoDB. A green notification banner at the top states "The Employee table was created successfully." Below this, the "Tables (1)" section displays a table named "Employee" with the following details:

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
Employee	Active	user_id (S)	name (S)	0	0	Off	☆	On-demand

Table created → select your table → explore table items → create items



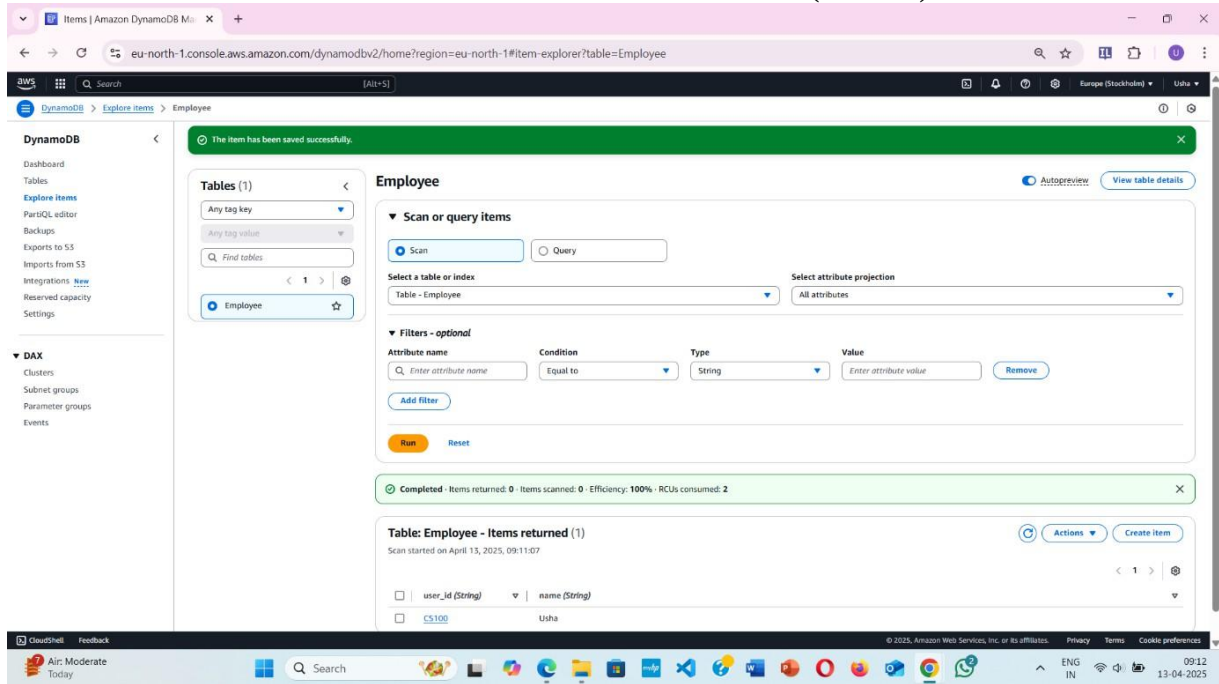
The screenshot shows the AWS Management Console for Amazon DynamoDB. The "Employee" table is selected, and the "Actions" dropdown menu is open, highlighting the "Explore items" option. The table details are the same as in the previous screenshot.



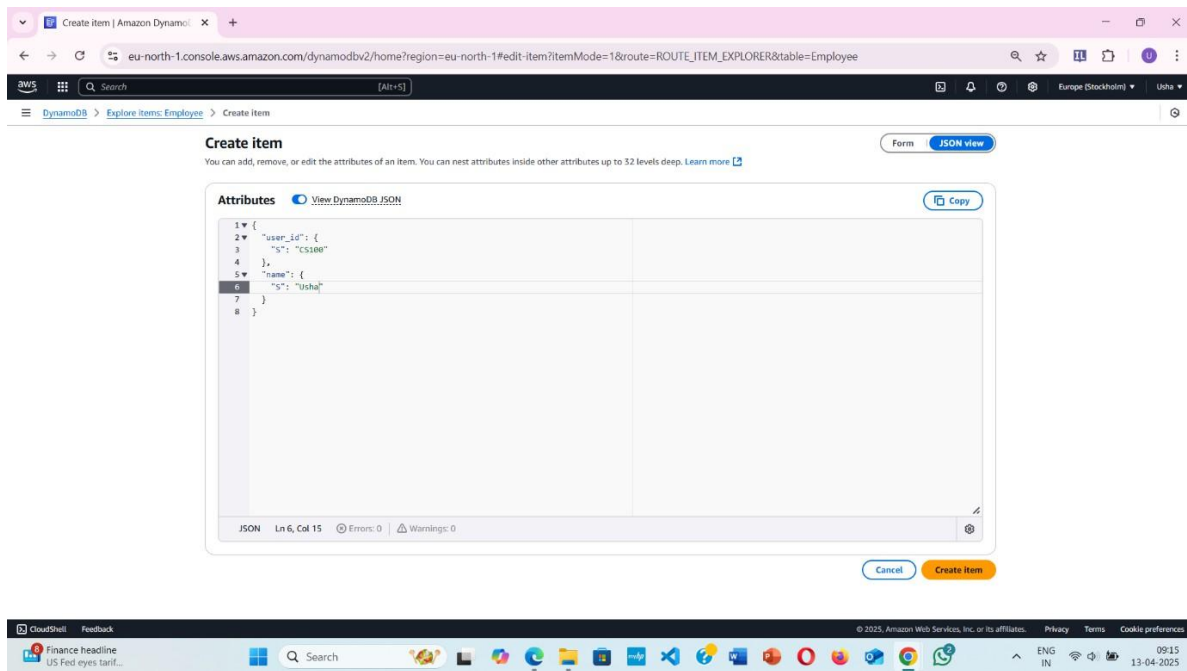
CLOUD COMPUTING LABORATORY (BCS601)

The screenshot shows the AWS DynamoDB console interface. The left sidebar contains navigation options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area displays the 'Employee' table explorer. It shows a 'Completed' status with 0 items returned and 100% efficiency. Below this, it states 'Table: Employee - Items returned (0)' and 'Scan started on April 13, 2025, 09:11:07'. A message indicates 'No items to display' with a 'Create item' button.

The screenshot shows the 'Create item' form in the AWS DynamoDB console. The form has a 'Form' tab and a 'JSON view' tab. It contains a table with two attributes: 'user_id' (Partition key) with value 'CS100' and 'name' (Sort key) with value 'Usha'. Both attributes are of type 'String'. There are 'Cancel' and 'Create item' buttons at the bottom.



Using JSON to create items





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The screenshot shows the AWS DynamoDB console interface. On the left, there's a navigation menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area displays the 'Employee' table. A message at the top says 'The item has been saved successfully.' Below this, there's a section for 'Scan or query items' with a 'Scan' button selected. The 'Select table or index' dropdown is set to 'Table - Employee', and 'Select attribute projection' is set to 'All attributes'. A 'Run' button is visible. Below the scan section, a green status bar indicates 'Completed - Items returned: 0 - Items scanned: 0 - Efficiency: 100% - RCUs consumed: 2'. The table view shows two items:

user_id (String)	name (String)
CS99	Usha
CS100	Usha

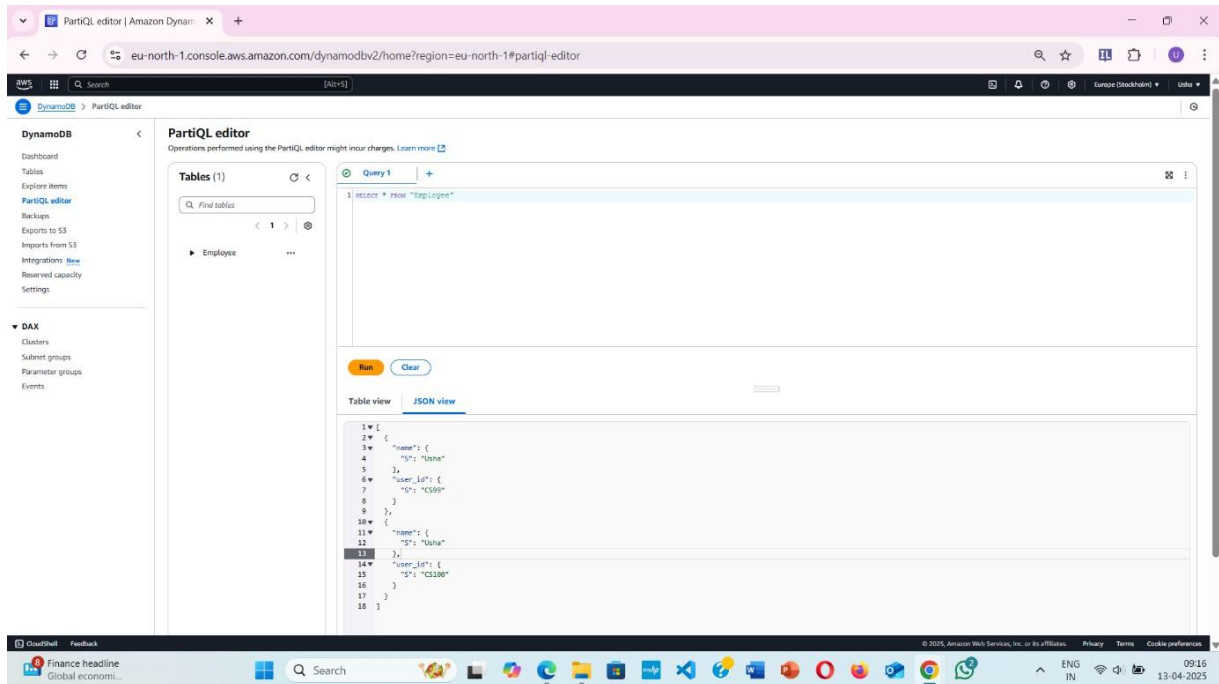
DynamoDB → PartiQL editor → write the queries

Table view

The screenshot shows the AWS PartiQL editor interface. The left sidebar has a navigation menu with options like Dashboard, Tables, Explore items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area displays the 'Employee' table. A query is entered in the 'Query 1' field: `SELECT * FROM Employee`. Below the query, there's a 'Run' button. The results are shown in a table view:

name	user_id
Usha	CS99
Usha	CS100

JSON view

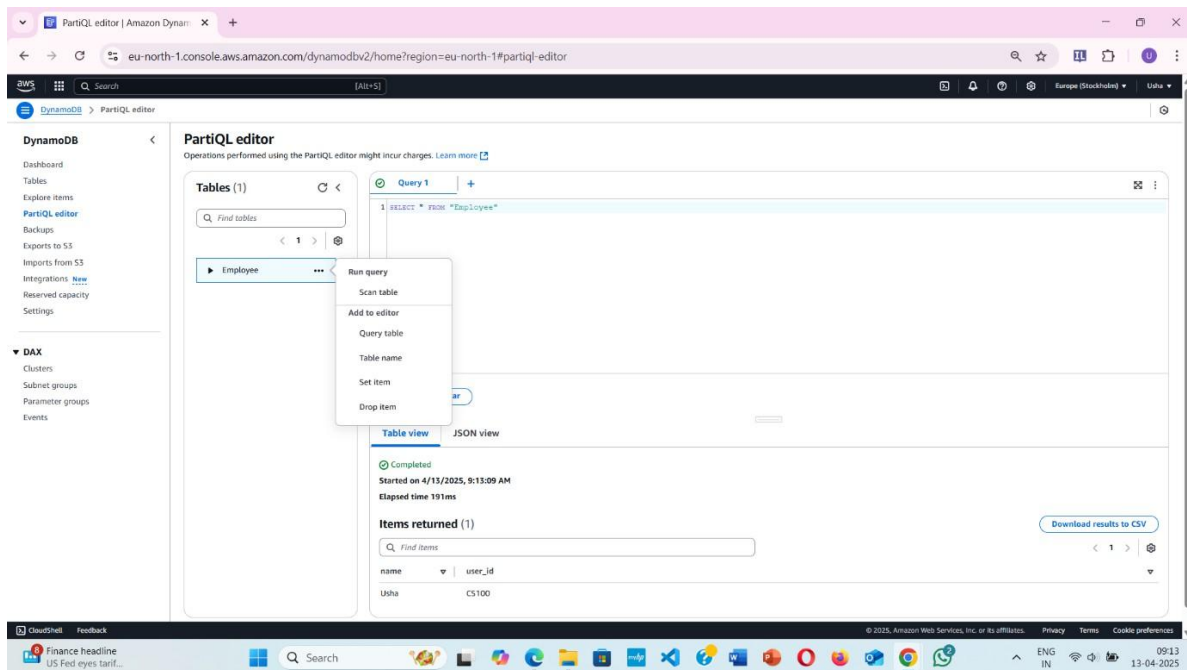


The screenshot shows the PartiQL editor interface in the AWS console. The left sidebar contains navigation links for DynamoDB, DAX, and other services. The main area displays a query editor with a query: `SELECT * FROM "Employee"`. Below the query editor, the results are shown in JSON view. The results are a list of items, each containing a `name` and a `user_id`.

```

1 [
2   {
3     "name": {
4       "S": "Usha"
5     },
6     "user_id": {
7       "S": "CS100"
8     }
9   },
10  {
11    "name": {
12      "S": "Usha"
13    },
14    "user_id": {
15      "S": "CS100"
16    }
17  }
18 ]
  
```

Update Command

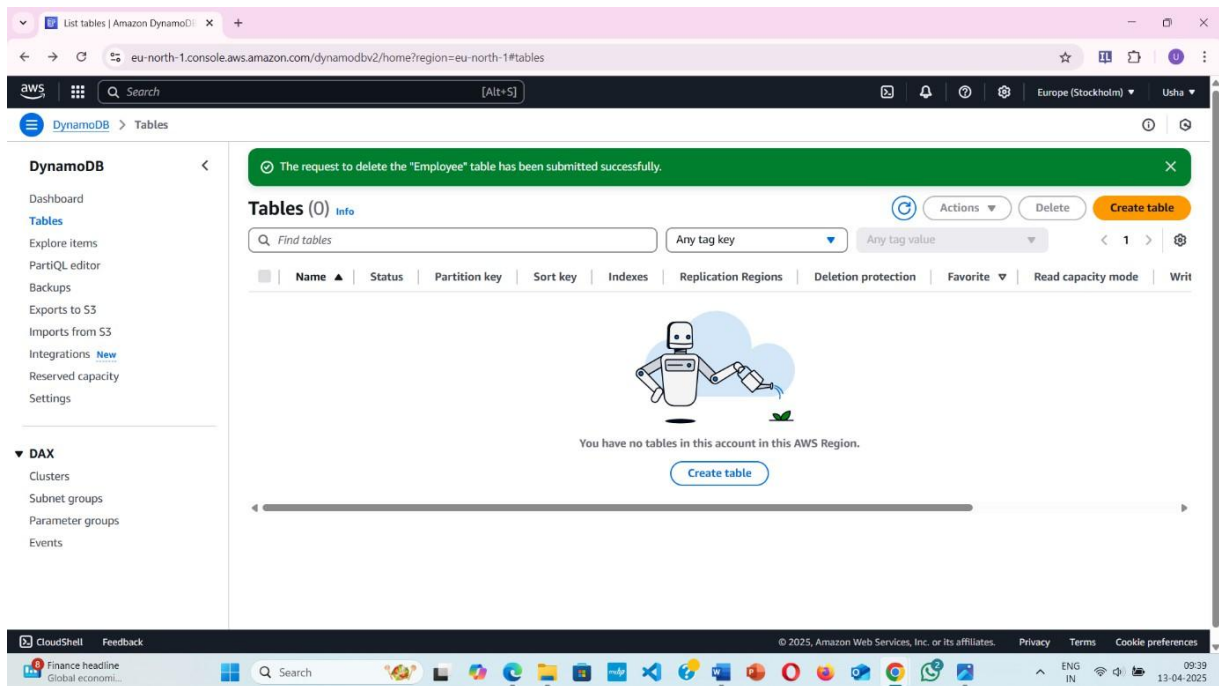
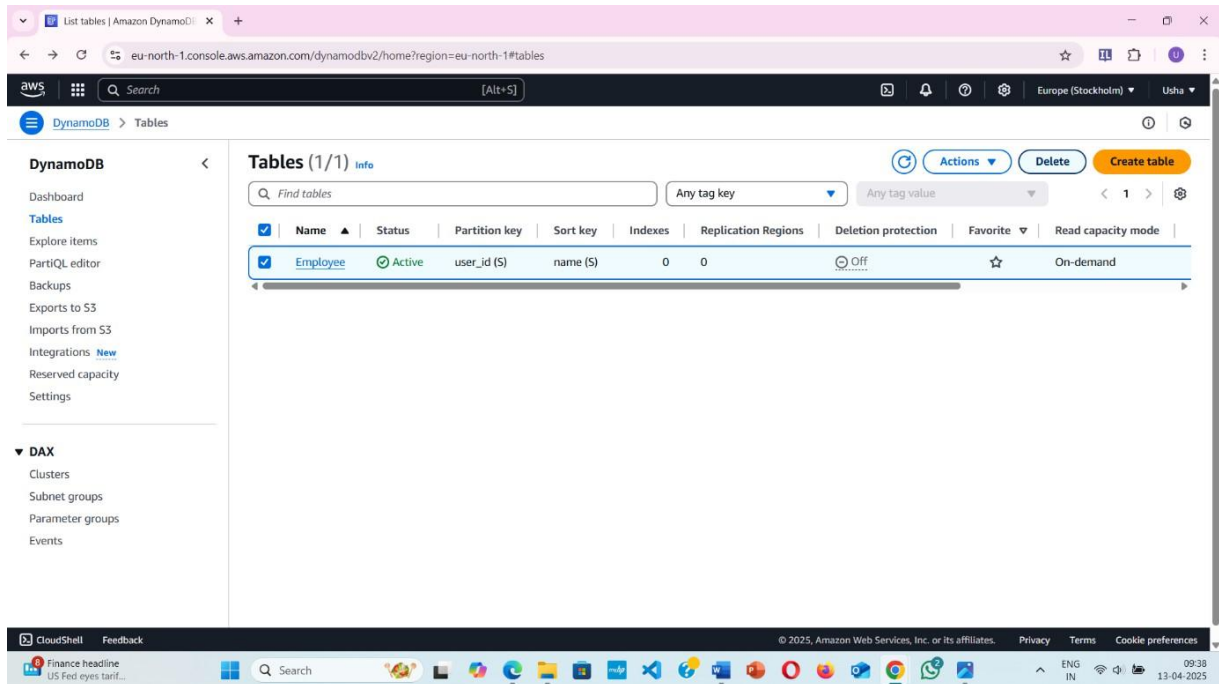


The screenshot shows the PartiQL editor interface in the AWS console. The left sidebar contains navigation links for DynamoDB, DAX, and other services. The main area displays a query editor with a query: `SELECT * FROM "Employee"`. Below the query editor, the results are shown in JSON view. The results are a list of items, each containing a `name` and a `user_id`.

```

1 [
2   {
3     "name": {
4       "S": "Usha"
5     },
6     "user_id": {
7       "S": "CS100"
8     }
9   },
10  {
11    "name": {
12      "S": "Usha"
13    },
14    "user_id": {
15      "S": "CS100"
16    }
17  }
18 ]
  
```


Delete table



Observations:

Tables and items are created, and we can run the commands to manipulate the data in the tables.

Experiment No – 04

Developing REST APIs with Amazon API Gateway

Objectives: Developing REST APIs with Amazon API Gateway.

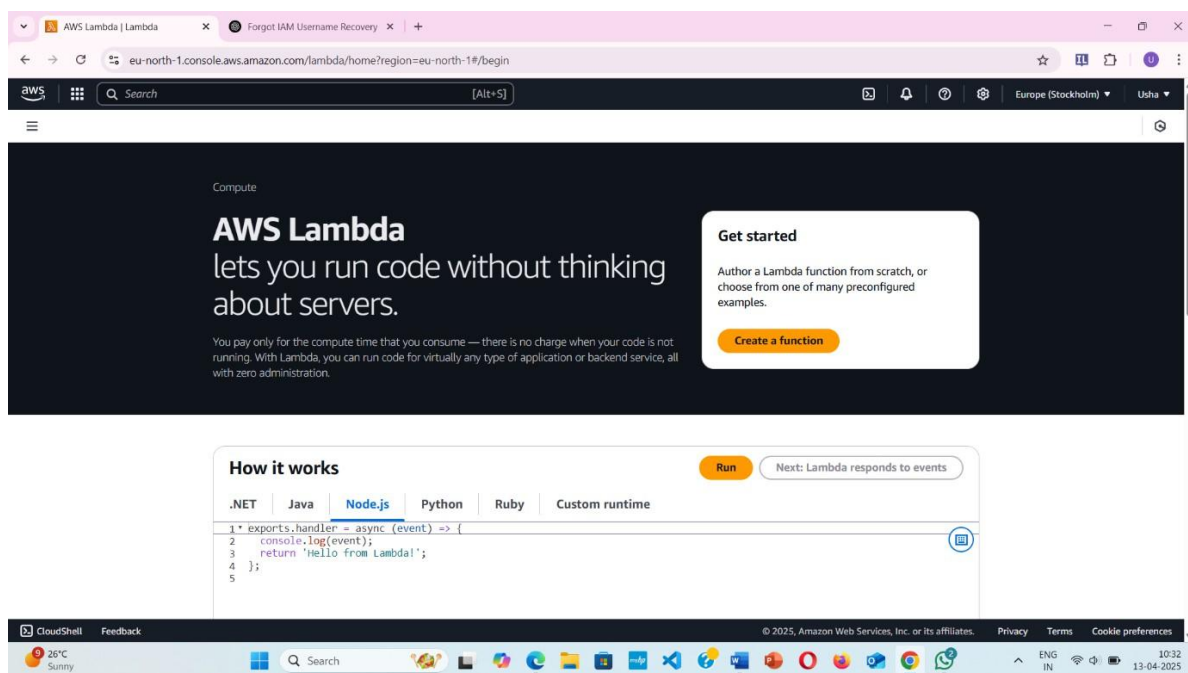
Apparatus Required: AWS Account

Pre-Requisite: Pre - requisite is creation of Lambda Function

Introduction: API Gateway enables you to connect and access data, business logic, and functionality from backend services such as workloads running on Amazon Elastic Compute Cloud (Amazon EC2), code running on AWS Lambda, any web application, or real-time communication applications.

Procedure

Open AWS Lambda





Create function

Create function [Info](#)

AWS Serverless Application Repository applications have moved to [Create application](#).

☒ **Author from scratch**
Start with a simple Hello World example.

☐ **Use a blueprint**
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphen (-), and underscore (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Architecture [Info](#)
Choose the instruction set architecture you want for your function code.
☒ x86_64
☐ arm64

Permissions [Info](#)
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

Change default execution role

Execution role
Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☒ Create a new role with basic Lambda permissions
☐ Use an existing role
☐ Create a new role from AWS policy templates

Role creation might take a few minutes. Please do not delete the role or edit the trust or permissions policies in this role.

Additional Configurations
Use additional configurations to set up code signing, function URL, tags, and Amazon VPC access for your function.

☐ **Enable code signing** [Info](#)
Use code signing configurations to ensure that the code has been signed by an approved source and has not been altered since signing.

☐ **Enable encryption with an AWS KMS customer managed key** [Info](#)
By default, Lambda encrypts the .jar file archive using an AWS owned key.

☒ **Enable function URL** [Info](#)
Use function URLs to assign HTTP(S) endpoints to your Lambda function.

Auth type
Choose the auth type for your function URL. [Learn more](#).

☐ AWS_IAM
Only authenticated IAM users and roles can make requests to your function URL.

☒ NONE
Lambda won't perform IAM authentication on requests to your function URL. The URL endpoint will be public unless you implement your own authorization logic in your function.

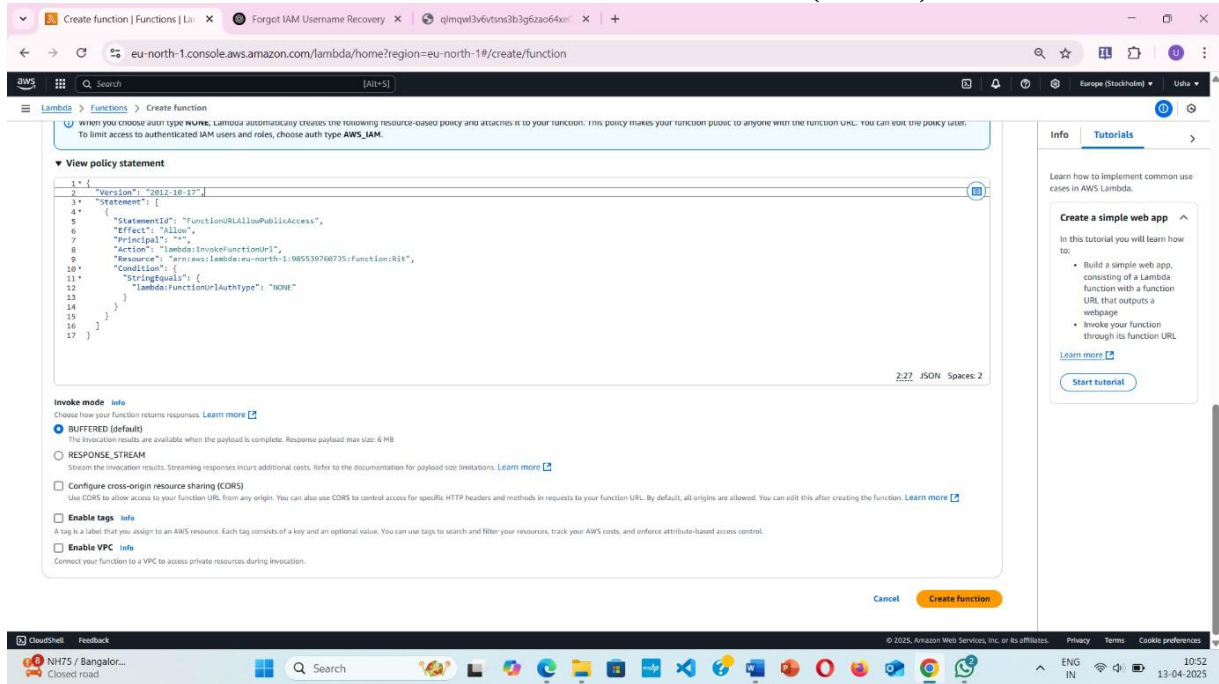
Function URL permissions

When you choose auth type **NONE**, Lambda automatically creates the following resource-based policy and attaches it to your function. This policy makes your function public to anyone with the function URL. You can edit the policy later. To limit access to authenticated IAM users and roles, choose auth type **AWS_IAM**.

View policy statement

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "StatementId": "FunctionURLAllowPublicAccess",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "lambda:InvokeFunctionUrl",
9       "Resource": "arn:aws:lambda:eu-north-1:985539760735:function:RIT",
10      "Condition": {
11        "StringEquals": {
12          "lambda:FunctionUrlAuthType": "NONE"
13        }
14      }
15    }
16  ]
17 }
```

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When you choose **NAME**, Lambda automatically creates the following resource-based policy and attaches it to your function. This policy makes your function public to anyone with the function URL. You can edit the policy later. To limit access to authenticated IAM users and roles, choose **auth type** **AWS_IAM**.

View policy statement

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "StatementId": "FunctionURLAllowPublicAccess",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "lambda:InvokeFunctionUrl",
9       "Resource": "arn:aws:lambda:eu-north-1:985539760735:function:Rit",
10      "Condition": {
11        "StringEquals": {
12          "lambda:FunctionUrlAuthType": "NONE"
13        }
14      }
15    }
16  ]
17 }
```

2/2 JSON Spaces: 2

Invoke mode info

Choose how your function returns responses. [Learn more](#)

☒ **BUFFERED (default)**
The invocation results are available when the payload is complete. Response payload max size: 6 MB

☐ **RESPONSE_STREAM**
Stream the invocation results. Streaming responses incurs additional costs. Refer to the documentation for payload size limitations. [Learn more](#)

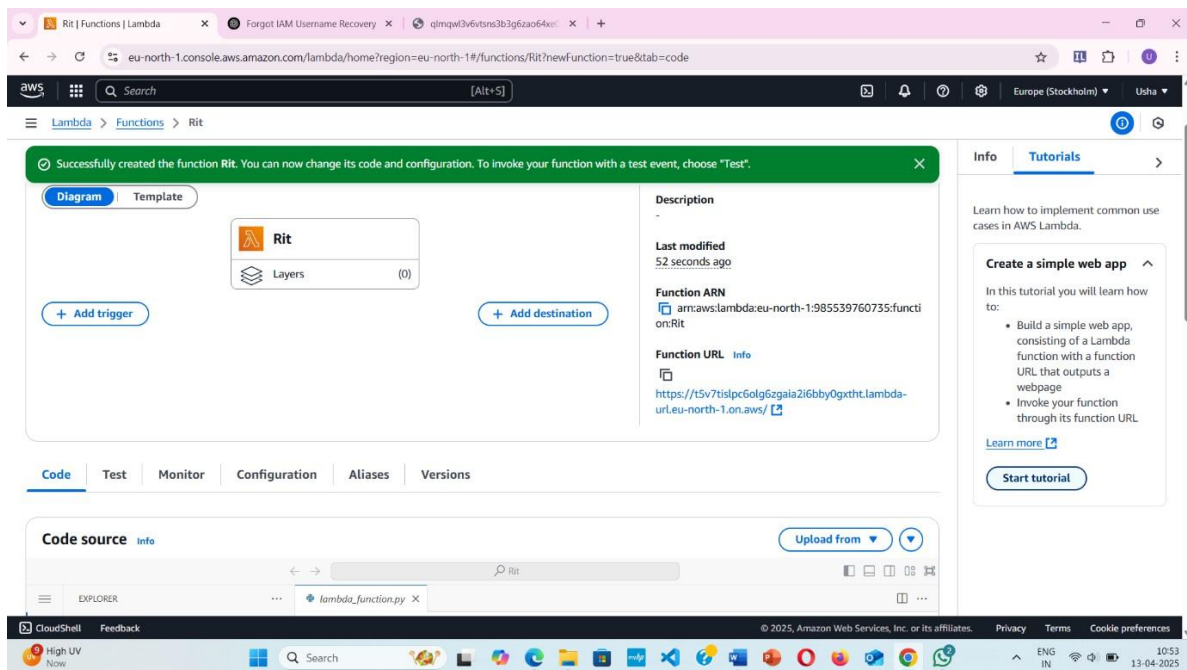
☐ **Configure cross-origin resource sharing (CORS)**
Use CORS to allow access to your function URL from any origin. You can also use CORS to control access for specific HTTP headers and methods in requests to your function URL. By default, all origins are allowed. You can edit this after creating the function. [Learn more](#)

☐ **Enable tags** info
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources, track your AWS costs, and enforce attribute-based access control.

☐ **Enable VPC** info
Connect your function to a VPC to access private resources during invocation.

[Cancel](#) [Create function](#)

Successfully created the function



Successfully created the function **Rit**. You can now change its code and configuration. To invoke your function with a test event, choose **"Test"**.

Diagram **Template**

Rit

Layers (0)

[+ Add trigger](#) [+ Add destination](#)

Description

Last modified 52 seconds ago

Function ARN
arn:aws:lambda:eu-north-1:985539760735:function:Rit

Function URL info
<https://t5v7tislpc60g6zgaia2f6bby0gxtht.lambda-url.eu-north-1.on.aws/>

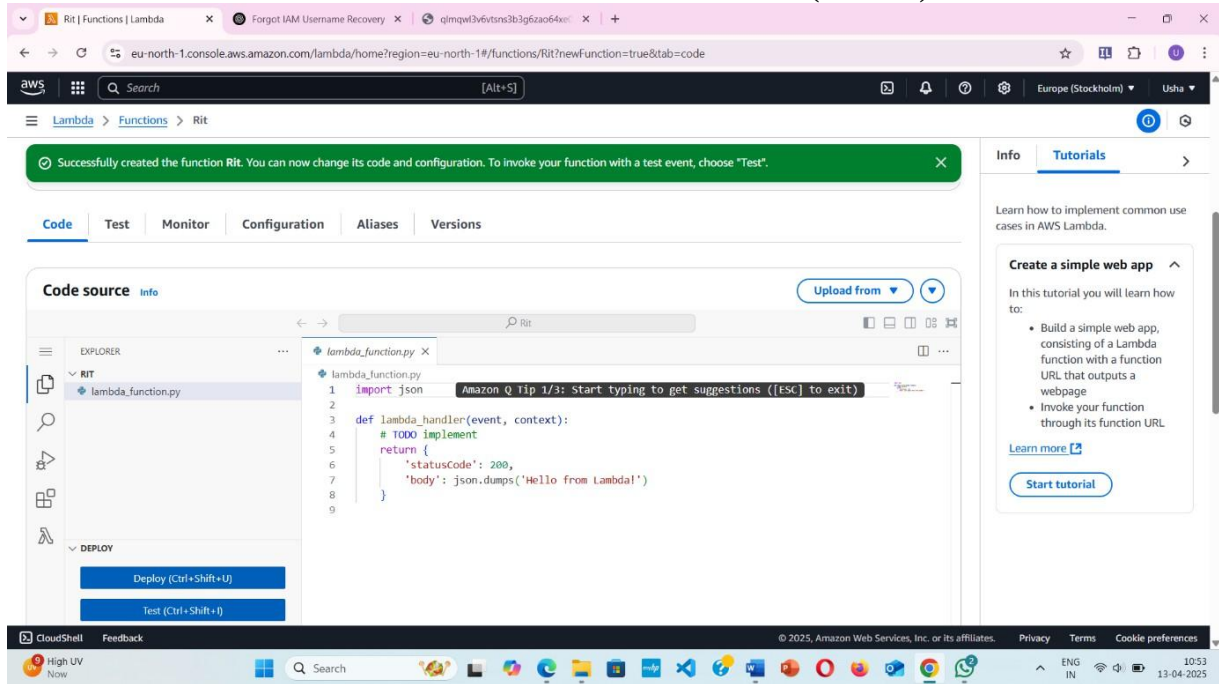
Code source info

[Upload from](#)

EXPLORER

lambda_function.py

CLOUD COMPUTING LABORATORY (BCS601)



Successfully created the function Rit. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

Code Test Monitor Configuration Aliases Versions

Code source Info

Upload from

EXPLORER

lambda_function.py

```

1 import json
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
7         'body': json.dumps('Hello from Lambda!')
8     }
9

```

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

CloudShell Feedback

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High UV Now

Search

ENG IN 10:53 13-04-2025

Info Tutorials

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

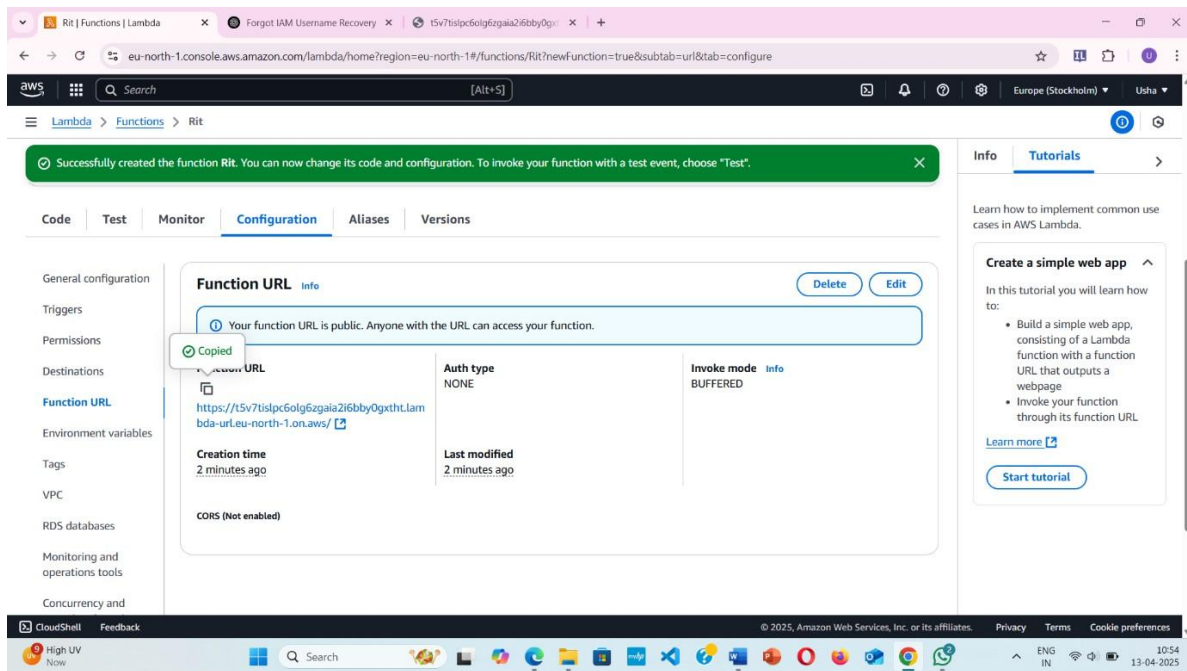
In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

Learn more

Start tutorial

Configuration → function URL → check URL runs



Successfully created the function Rit. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

Code Test Monitor **Configuration** Aliases Versions

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

RDS databases

Monitoring and operations tools

Concurrency and

Function URL Info

Your function URL is public. Anyone with the URL can access your function.

Copy

Function URL

https://t5v7tislpc6olg6zgaia2i6bby0gxthLam bda-url.eu-north-1.on.aws/

Auth type

NONE

Invoke mode

Info

BUFFERED

Creation time

2 minutes ago

Last modified

2 minutes ago

CORS (Not enabled)

CloudShell Feedback

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Info Tutorials

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

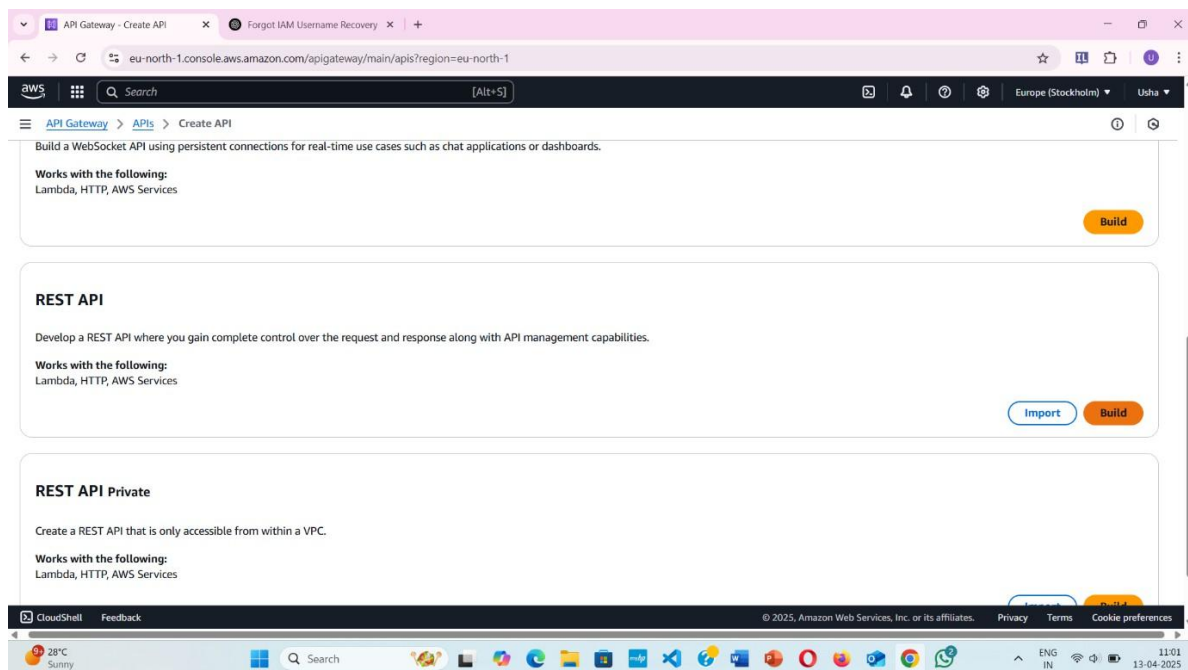
Learn more

Start tutorial

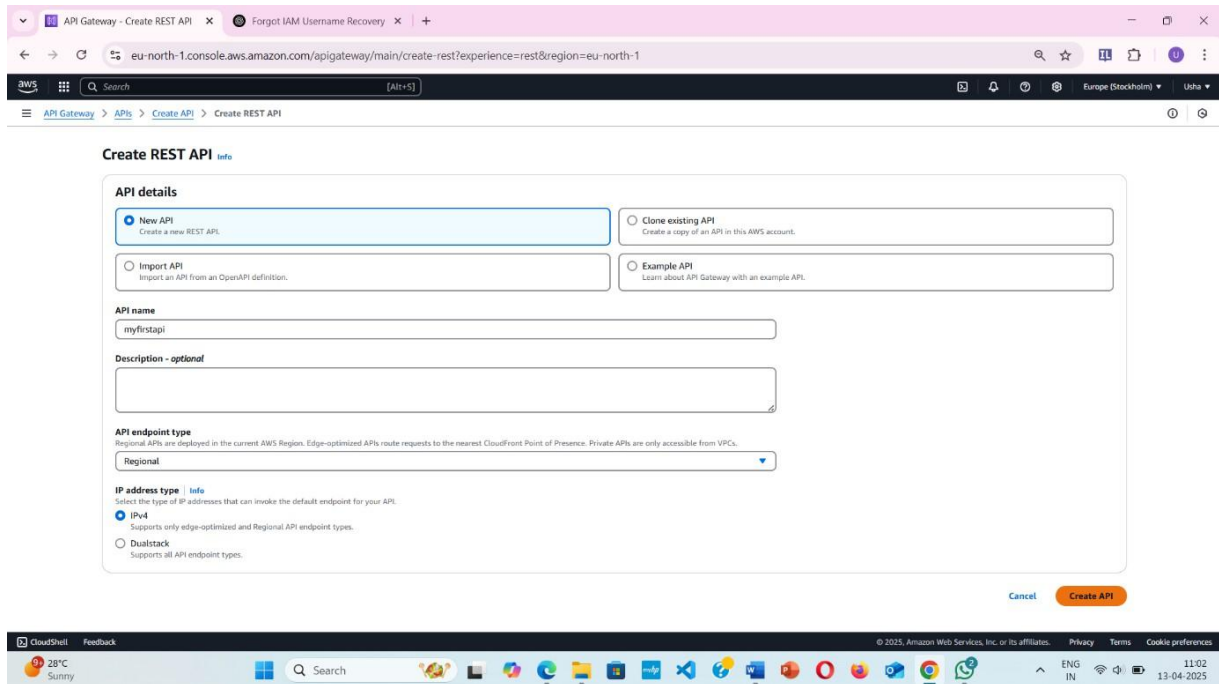
Lambda Code Samples

```
import json
print(json.dumps({"name": "John", "age": 30}))
print(json.dumps(["apple", "banan as"]))
print(json.dumps(("apple", "bananas")))
print(json.dumps("hello"))
print(json.dumps(42))
print(json.dumps(31.76))
print(json.dumps(True))
print(json.dumps(False))
print(json.dumps(None))
```

Go to API Gateway in Console And Select Rest API



Build Rest API



Create REST API Info

API details

☒ **New API**
Create a new REST API.

☐ **Clone existing API**
Create a copy of an API in this AWS account.

☐ **Import API**
Import an API from an OpenAPI definition.

☐ **Example API**
Learn about API Gateway with an example API.

API name
myfirstapi

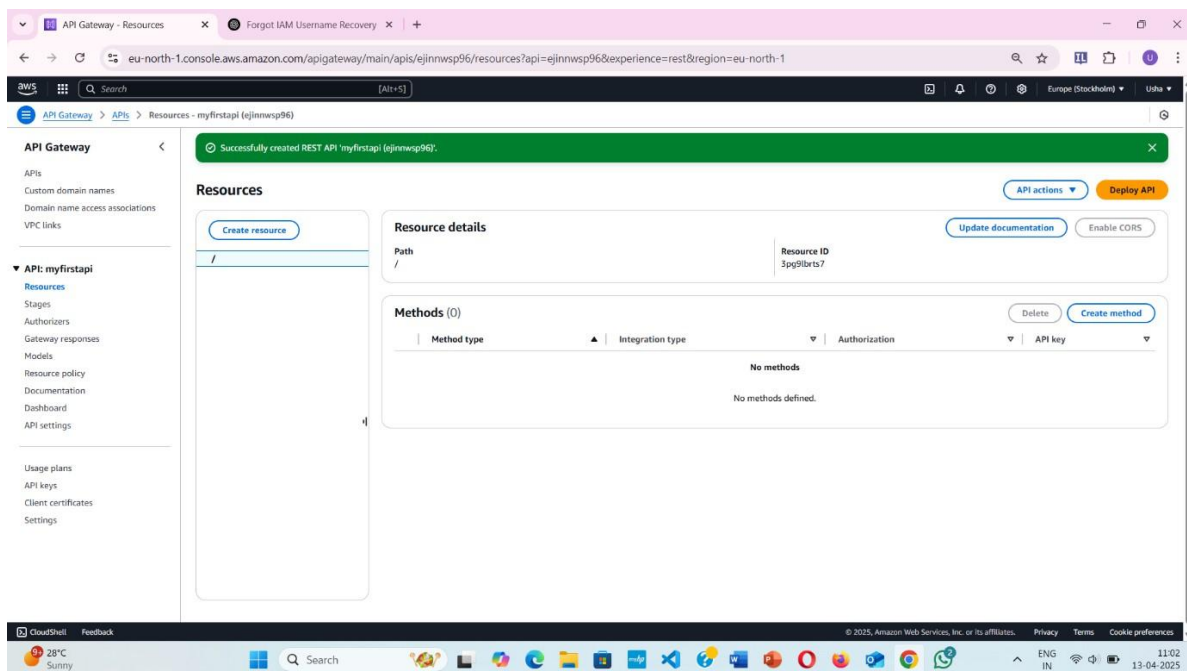
Description - optional

API endpoint type
Regional
Regional APIs are deployed in the current AWS Region. Edge-optimized APIs route requests to the nearest CloudFront Point of Presence. Private APIs are only accessible from VPCs.

IP address type Info
Select the type of IP addresses that can invoke the default endpoint for your API.
☒ **IPv4**
Supports only edge-optimized and Regional API endpoint types.
☐ **Dualstack**
Supports all API endpoint types.

[Cancel](#) [Create API](#)

Successfully created the REST API



API Gateway Resources

Resources

[Create resource](#)

Resource details

Path: /

Resource ID: 3pg9lbrs7

[API actions](#) [Deploy API](#) [Update documentation](#) [Enable CORS](#)

Methods (0)

[Delete](#) [Create method](#)

Method type	Integration type	Authorization	API key
No methods defined.			

Create method And Select Lambda Integration

Select GET method

Successfully created REST API 'myfirstapi (ejinnwsp96)'.

Create method

Method details

Method type: GET

Integration type

☒ Lambda function
Integrate your API with a Lambda function.

☐ HTTP
Integrate with an existing HTTP endpoint.

☐ Mock
Generate a response based on API Gateway mappings and transformations.

☐ AWS service
Integrate with an AWS Service.

☐ VPC link
Integrate with a resource that isn't accessible over the public internet.

☐ Lambda proxy integration
Send the request to your Lambda function as a structured event.

Lambda function
Provide the Lambda function name or alias. You can also provide an ARN from another account.

eu-north-1

Grant API Gateway permission to invoke your Lambda function. To turn off, update the function's resource policy yourself, or provide an invoke role that API Gateway uses to invoke your function.

Integration timeout Info
By default, you can enter an integration timeout of 50 - 29,000 milliseconds. You can use Service Quotas to raise the integration timeout to greater than 29,000 ms.

29000

Method request settings

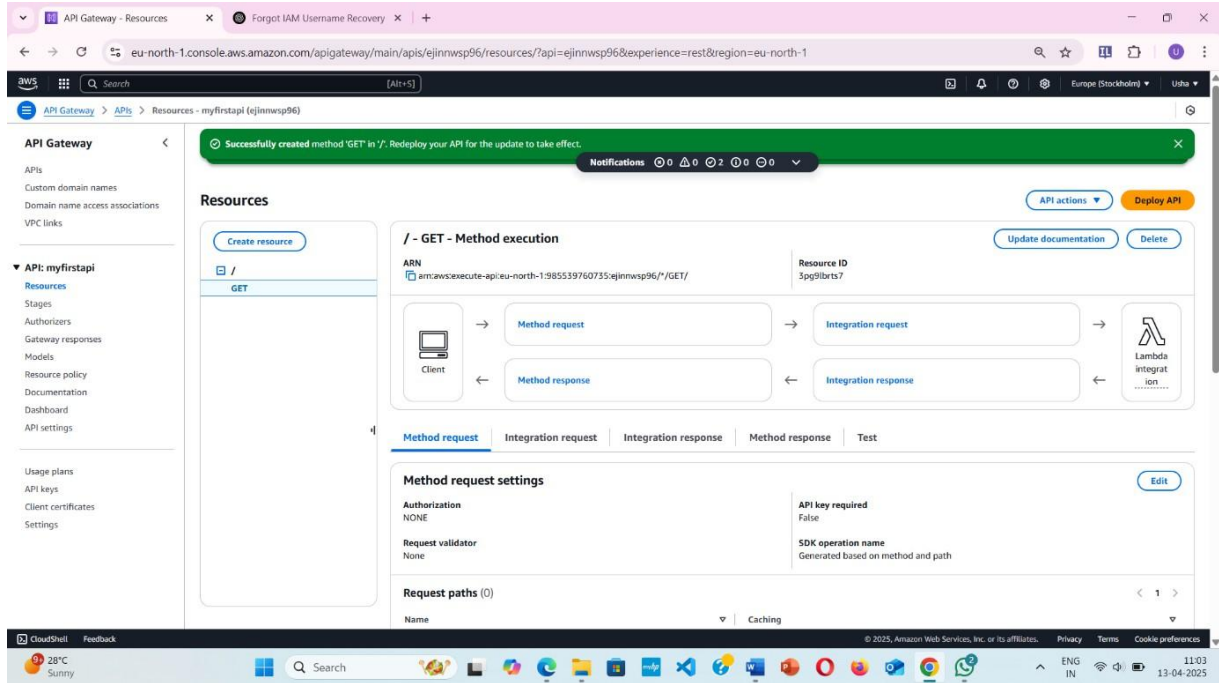
URL query string parameters

HTTP request headers

Request body

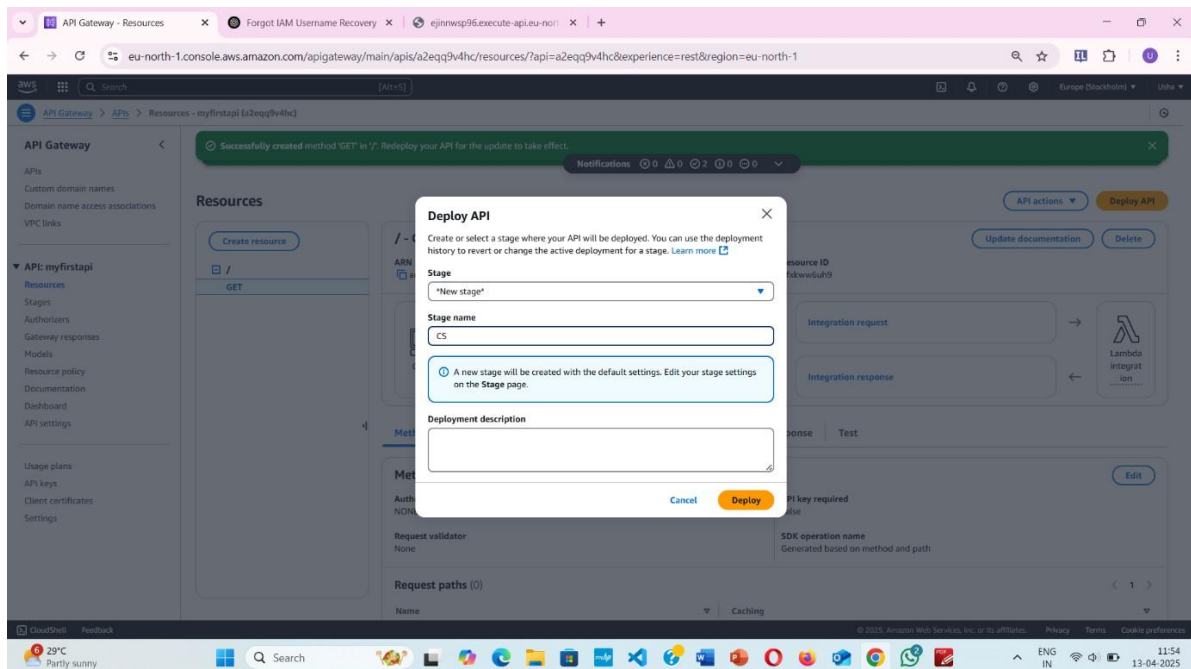
Cancel Create method

Successfully created method “GET”

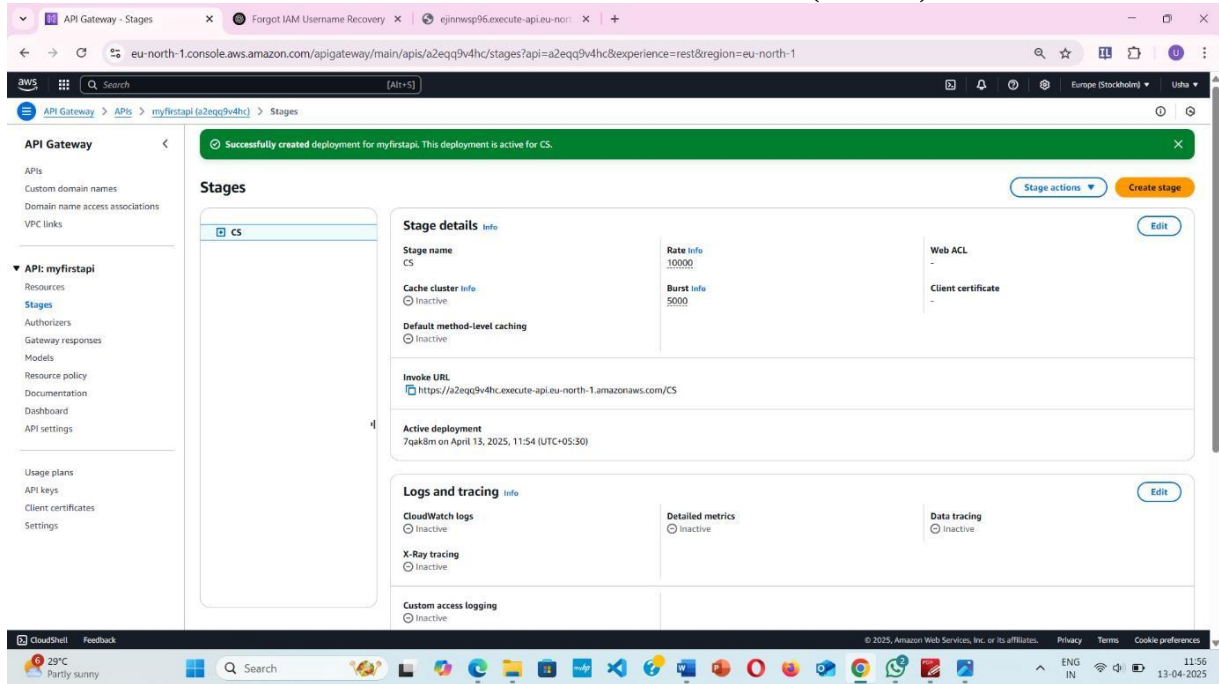


To Test API

Deploy API



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API Gateway - Stages

eu-north-1.console.aws.amazon.com/apigateway/main/apis/a2eqq9v4hc/stages?api=a2eqq9v4hc&experience=rest®ion=eu-north-1

API Gateway > APIs > myfirstapi (a2eqq9v4hc) > Stages

APIs
Custom domain names
Domain name access associations
VPC links

▼ API: myfirstapi
Resources
Stages
Authorizers
Gateway responses
Models
Resource policy
Documentation
Dashboard
API settings

Usage plans
API keys
Client certificates
Settings

Successfully created deployment for myfirstapi. This deployment is active for CS.

Stage actions Create stage

Stages

CS

Stage details

Stage name CS

Cache cluster Inactive

Default method-level caching Inactive

Invoke URL https://a2eqq9v4hc.execute-api.eu-north-1.amazonaws.com/CS

Active deployment 7qak8m on April 13, 2025, 11:54 (UTC+05:30)

Web ACL -

Client certificate -

Logs and tracing

CloudWatch logs Inactive

X-Ray tracing Inactive

Custom access logging Inactive

Detailed metrics Inactive

Data tracing Inactive

CloudShell Feedback

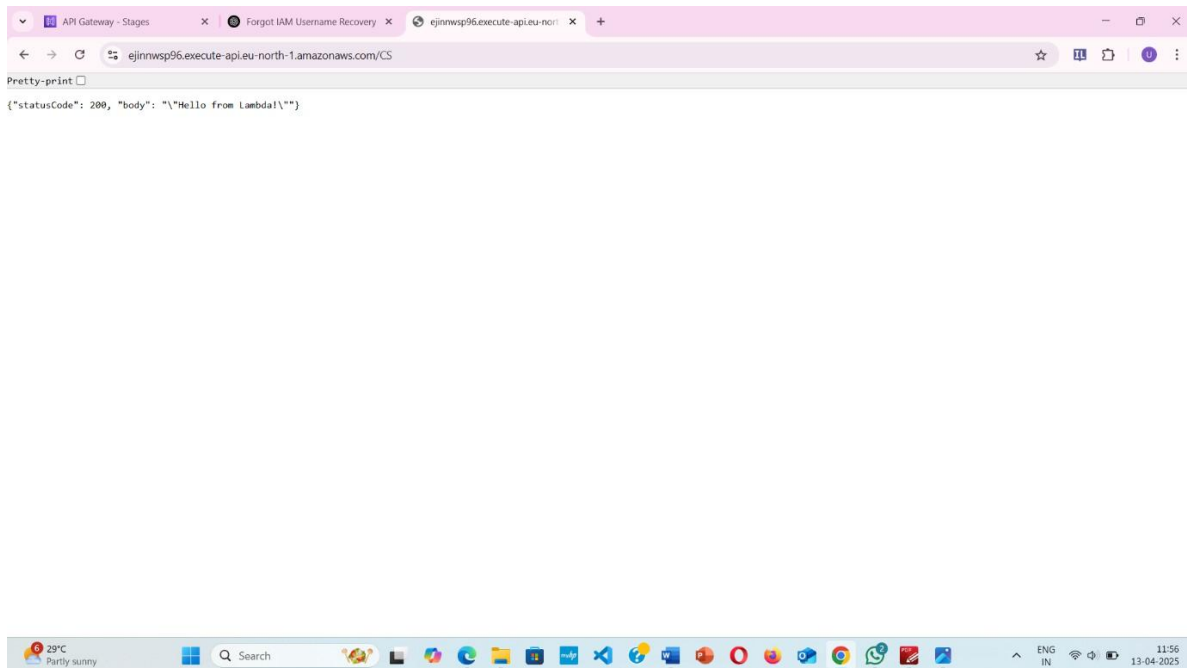
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29°C Partly sunny

Search

ENG IN 11:56 13-04-2025

Use URL and copy in new tab



API Gateway - Stages

Forgot IAM Username Recovery

ejinnwsp96.execute-api.eu-north-1.amazonaws.com/CS

Pretty-print

```
{
  "statusCode": 200,
  "body": "\"Hello from Lambda!\""
}
```

29°C Partly sunny

Search

ENG IN 11:56 13-04-2025

Experiment No – 5

Creating Lambda Functions Using the AWS SDK for Python

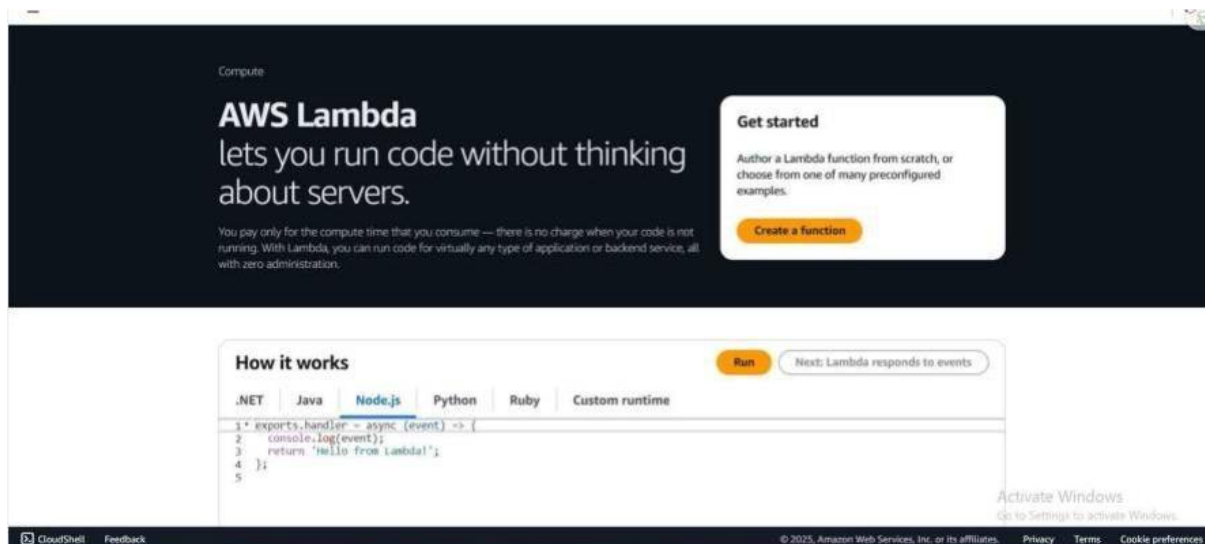
Objectives: Creating Lambda Functions Using the AWS SDK for Python

Apparatus Required: AWS Account

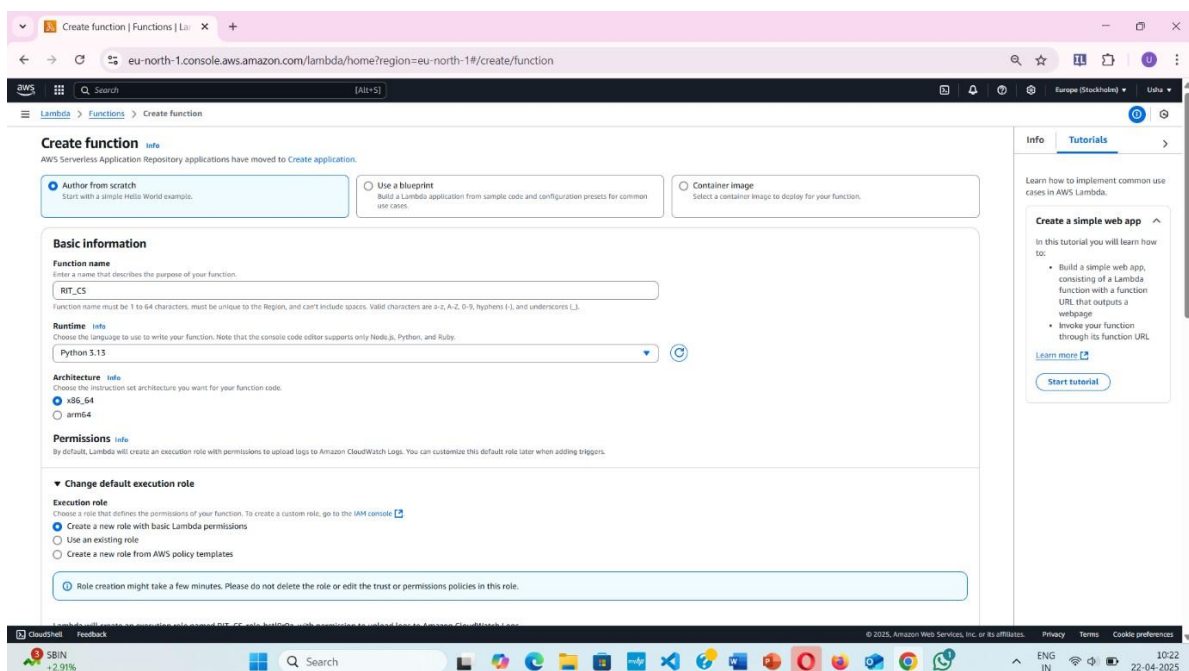
Pre-Requisite: AWS basic knowledge, basics of lambda functions and Python

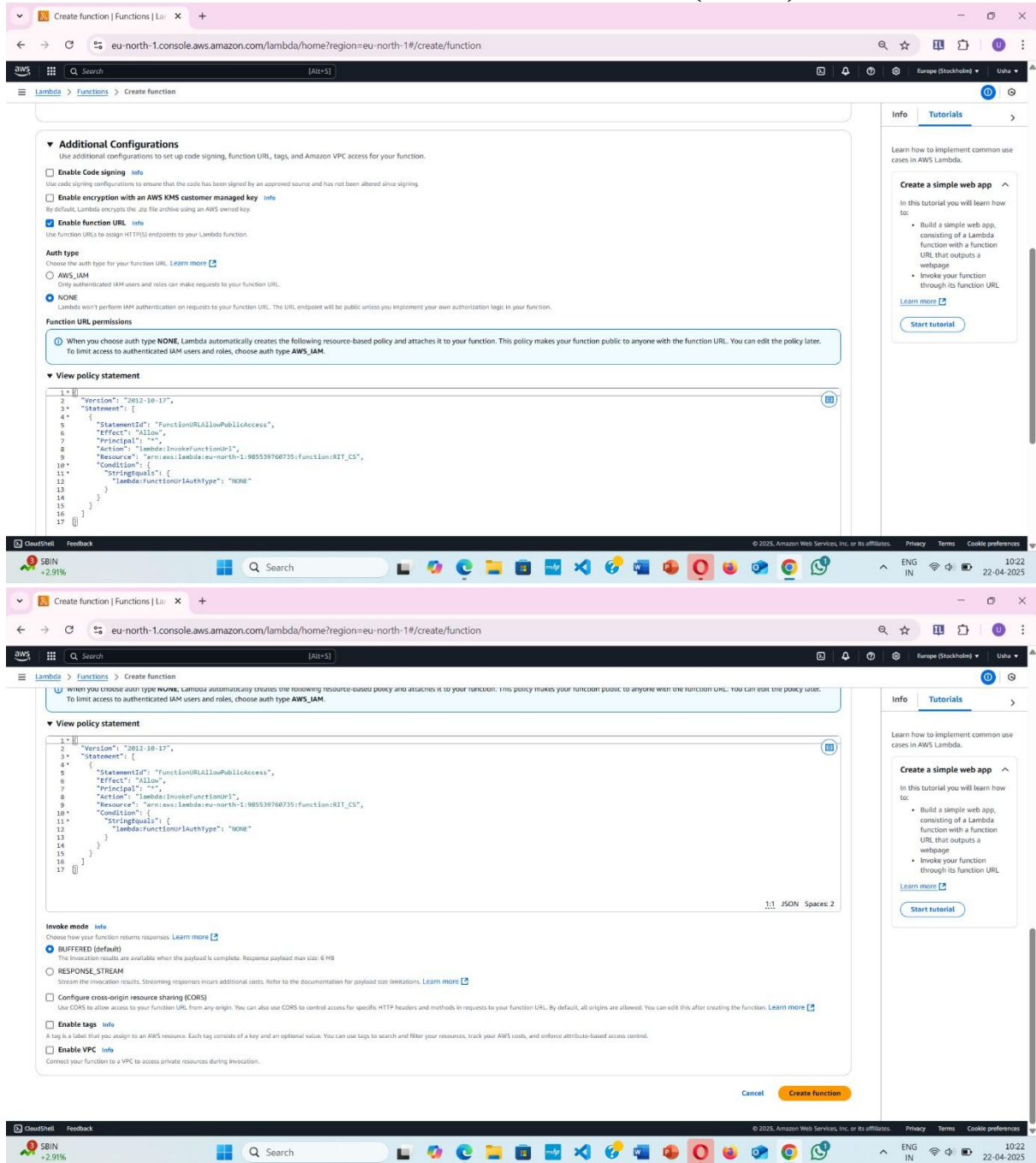
Introduction: An AWS Lambda function is a serverless function that runs code in response to events, such as user actions or system updates. You can use Lambda functions to create custom backend services, extend other AWS services, or trigger actions in other services.

Procedure:



CREATE FUNCTION





Additional Configurations
Use additional configurations to set up code signing, function URL, tags, and Amazon VPC access for your function.

- ☐ **Enable Code signing** [info](#)
Use code signing configurations to ensure that the code has been signed by an approved source and has not been altered since signing.
- ☐ **Enable encryption with an AWS KMS customer managed key** [info](#)
By default, Lambda encrypts the .zip file archive using an AWS owned key.
- ☒ **Enable function URL** [info](#)
Use function URLs to assign HTTPS endpoints to your Lambda function.

Auth type
Choose the auth type for your function URL. [Learn more](#)

- ☐ **AWS_IAM**
Only authenticated IAM users and roles can make requests to your function URL.
- ☒ **NONE**
Lambda won't perform IAM authentication on requests to your function URL. The URL endpoint will be public unless you implement your own authorization logic in your function.

Function URL permissions
When you choose auth type **NONE**, Lambda automatically creates the following resource-based policy and attaches it to your function. This policy makes your function public to anyone with the function URL. You can edit the policy later. To limit access to authenticated IAM users and roles, choose auth type **AWS_IAM**.

View policy statement

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "StatementId": "FunctionURLAllowPublicAccess",
6       "Effect": "Allow",
7       "Principal": "*",
8       "Action": "lambda:InvokeFunctionUrl",
9       "Resource": "arn:aws:lambda:eu-north-1:985539760735:function:RIT_CS",
10      "Condition": {
11        "StringEquals": {
12          "lambda:FunctionUrlAuthType": "NONE"
13        }
14      }
15    }
16  ]
17 }
```

Invoke mode [info](#)
Choose how your function returns responses. [Learn more](#)

- ☒ **BUFFERED (default)**
The invocation results are available when the payload is complete. Response payload max size: 6 MB
- ☐ **RESPONSE_STREAM**
Stream the invocation results. Streaming responses incurs additional costs. Refer to the documentation for payload size limitations. [Learn more](#)

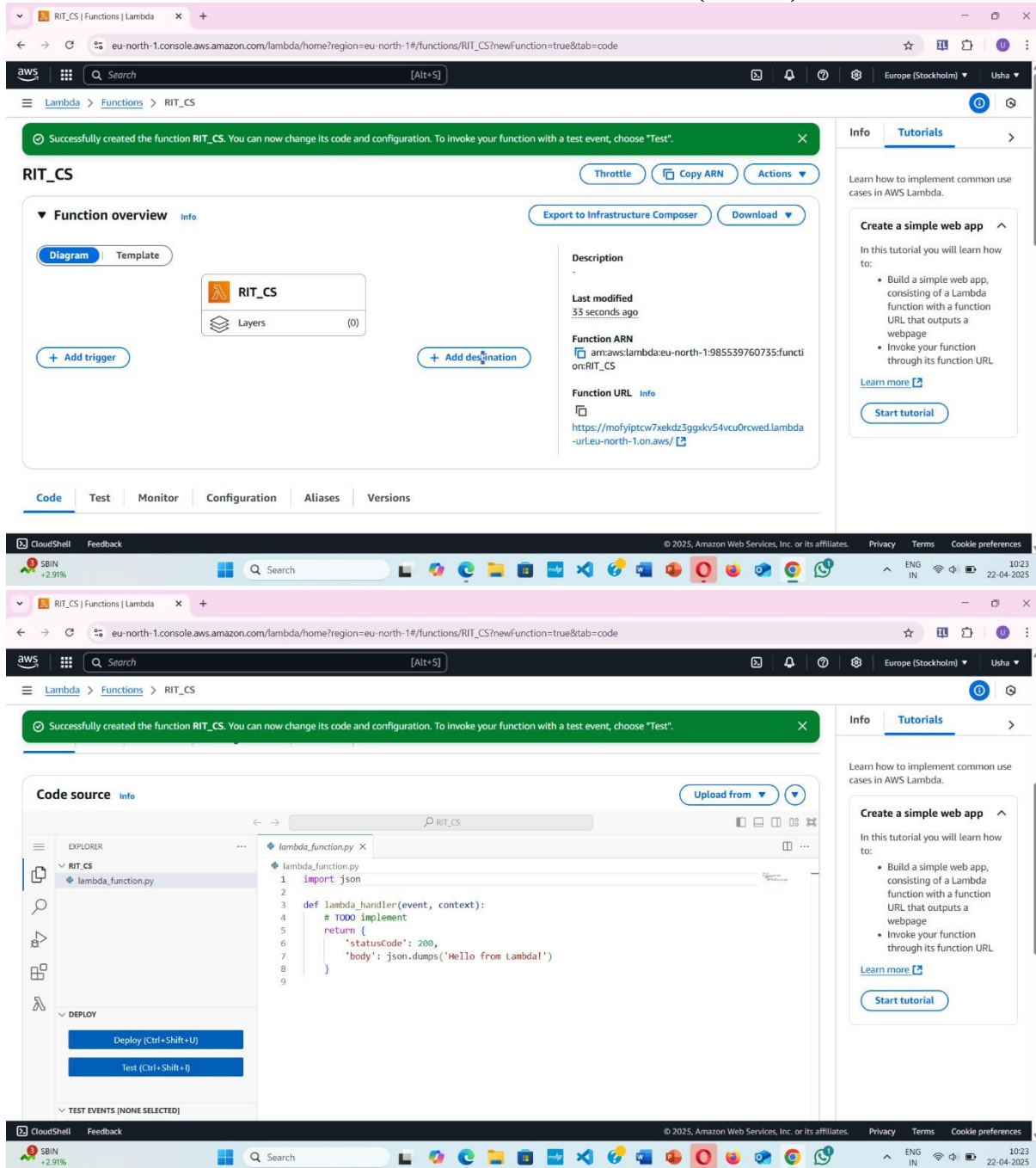
☐ **Configure cross-origin resource sharing (CORS)**
Use CORS to allow access to your function URL from any origin. You can also use CORS to control access for specific HTTP headers and methods in requests to your function URL. By default, all origins are allowed. You can edit this after creating the function. [Learn more](#)

☐ **Enable tags** [info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources, track your AWS costs, and enforce attribute-based access control.

☐ **Enable VPC** [info](#)
Connect your function to a VPC to access private resources during invocation.

[Cancel](#) [Create function](#)

CLOUD COMPUTING LABORATORY (BCS601)



The screenshot displays the AWS Lambda console interface. At the top, a green banner confirms the successful creation of the function 'RIT_CS'. The main content area is divided into two tabs: 'Function overview' and 'Code source'.

Function overview tab:

- Diagram:** Shows the function 'RIT_CS' with a 'Layers' section indicating 0 layers.
- Description:**
 - Last modified:** 33 seconds ago
 - Function ARN:** arn:aws:lambda:eu-north-1:985539760735:func:RIT_CS
 - Function URL:** https://mofyiptcw7ekdz3ggekvs4vcu0rcwed.lambda-url.eu-north-1.on.aws/

Code source tab:

- EXPLORER:** Shows the file structure with 'RIT_CS' and 'lambda_function.py'.
- Code editor:** Displays the Python code for 'lambda_function.py':

```

1 import json
2
3 def lambda_handler(event, context):
4     # TODO implement
5     return {
6         'statusCode': 200,
7         'body': json.dumps('Hello from Lambda!')
8     }
9

```
- DEPLOY:** Includes buttons for 'Deploy (Ctrl+Shift+U)' and 'Test (Ctrl+Shift+I)'.
- TEST EVENTS:** Currently shows 'NONE SELECTED'.

The right sidebar contains a 'Tutorials' section with a link to 'Create a simple web app'.



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CONFIGERATIONS →FUNCTION URL → CHECK URL RUNS

The screenshot displays the AWS Lambda console for a function named 'RIT_CS'. A green notification banner at the top states: 'Successfully created the function RIT_CS. You can now change its code and configuration. To invoke your function with a test event, choose "Test".' The 'Configuration' tab is selected, showing the 'Function URL' section. A message indicates: 'Your function URL is public. Anyone with the URL can access your function.' The function URL is displayed as 'https://mofyiptcw7xekdz3ggxkv54vcu0rcwed.lambda-url.eu-north-1.on.aws/'. Other details include 'Auth type: NONE', 'Invoke mode: BUFFERED', 'Creation time: 1 minute ago', and 'Last modified: 1 minute ago'. A 'CORS (Not enabled)' section is also visible. On the right, a tutorial for 'Create a simple web app' is shown. Below the console, a terminal window shows the command 'curl https://mofyiptcw7xekdz3ggxkv54vcu0rcwed.lambda-url.eu-north-1.on.aws' and the output 'Hello from Lambda!'.

Experiment No – 06**Migrating a Web Application to Docker Containers****Objectives:** Migrating a Web Application to Docker Containers**Apparatus Required:** AWS Account**Pre-Requisite:** AWS basics knowledge, Basics of docker**Introduction:** Docker is a set of platform as a service (PaaS) products that use OS-level virtualization to deliver software in packages called containers.

Docker is a software platform that packages software into containers.

Docker images are read-only templates that contain instructions for creating a Container.

A Docker image is a snapshot or blueprint of the libraries and dependencies required inside a container for an application to run.

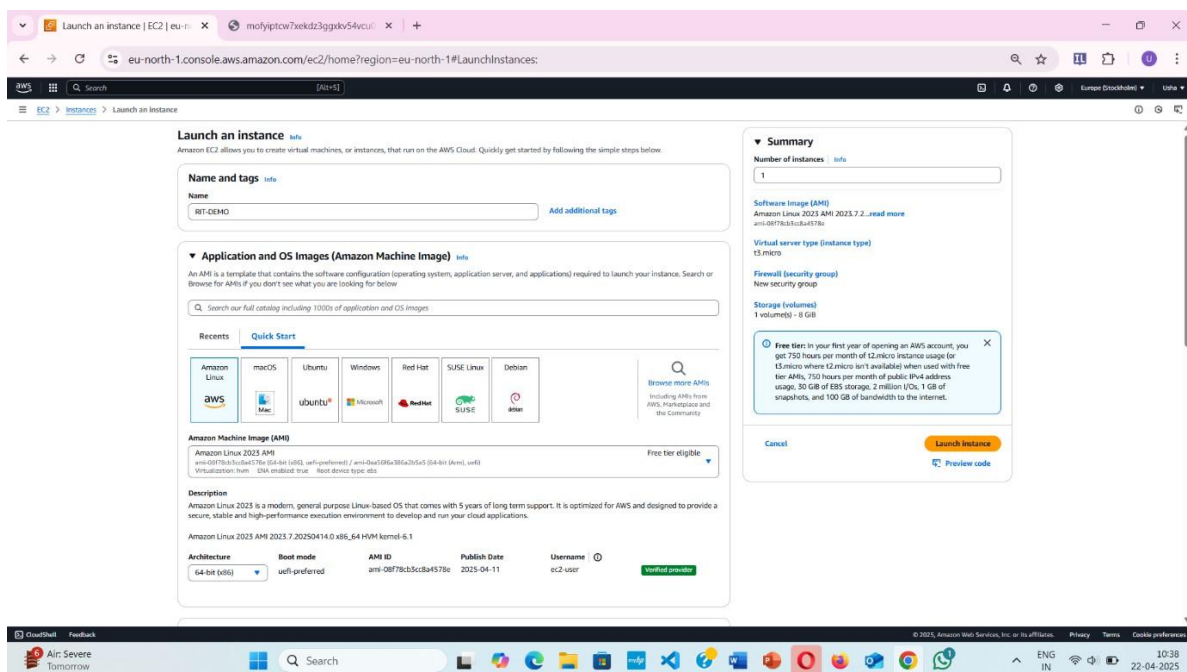
A Docker container image is a lightweight, standalone, executable package of software that includes everything needed to run an application: code, runtime,

Procedure:

Create EC2 Instance

Linux Machine

Enable HTTPS and H TTP





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Instance type [info](#) [Get advice](#)

Instance type: **t3.micro** Family: t3, 2 vCPU, 1 GiB Memory, Current generation instance, Free tier eligible [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key pair (login) [info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required: **Proceed without a key pair (Not recommended)** [Default value](#) [Create new key pair](#)

Network settings [info](#) [Edit](#)

Network: **vpc-08bf8afabce0ef593**

Subnet: **No preference (Default subnet in any availability zone)**

Auto-assign public IP: **Enable** [info](#)

Additional charges apply when outside of free tier allowance

Firewall (security groups) [info](#)

A security group is a set of firewall rules that control the traffic to and from your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called **launch-wizard-12** with the following rules:

- ☒ Allow SSH traffic from **Anywhere (0.0.0.0/0)** [info](#)
- ☒ Allow HTTPS traffic from the internet [info](#)
- ☒ Allow HTTP traffic from the internet [info](#)

Summary [info](#)

Number of instances: **1**

Software image (AMI): **Amazon Linux 2023 AMI 2023.7.2...read more** [em-001f8c0d0d0d0d0d0d](#)

Virtual server type (instance type): **t3.micro**

Firewall (security group): **New security group**

Storage (volumes): **1 volume(s) - 8 GiB**

[Cancel](#) [Launch instance](#) [Preview code](#)

Success

Successfully initiated launch of instance (i-00318b1e6db1f6b0a)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

[Create billing alerts](#)

Connect to your instance

Once your instance is running, log into it from your local computer.

[Connect to instance](#) [Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

[Connect an RDS database](#) [Create a new RDS database](#) [Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots.

[Create EBS snapshot policy](#)

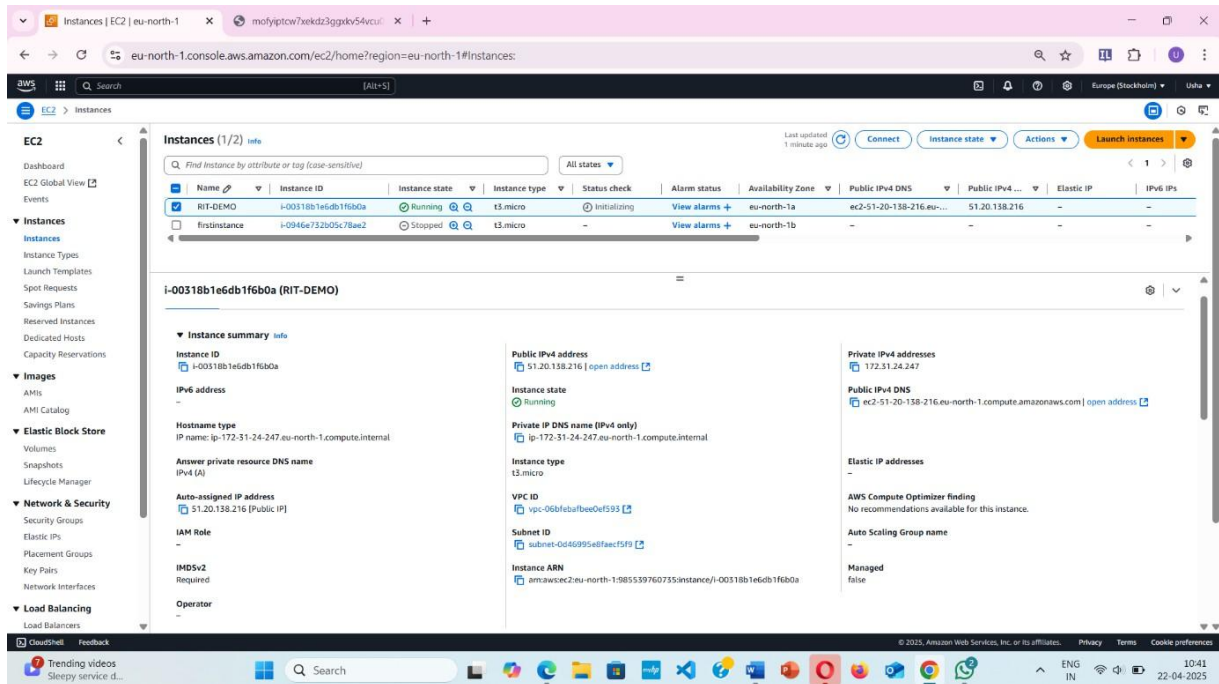
Manage detailed monitoring

Create Load Balancer

Create AWS budget

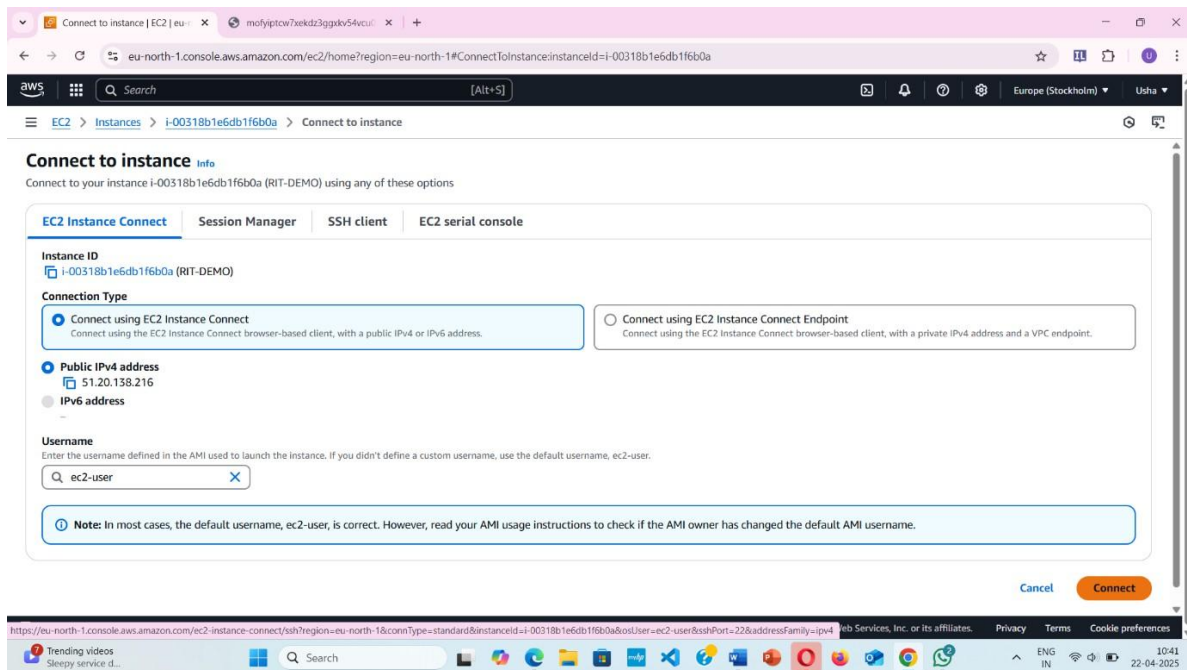
Manage CloudWatch alarms

Go To Instance



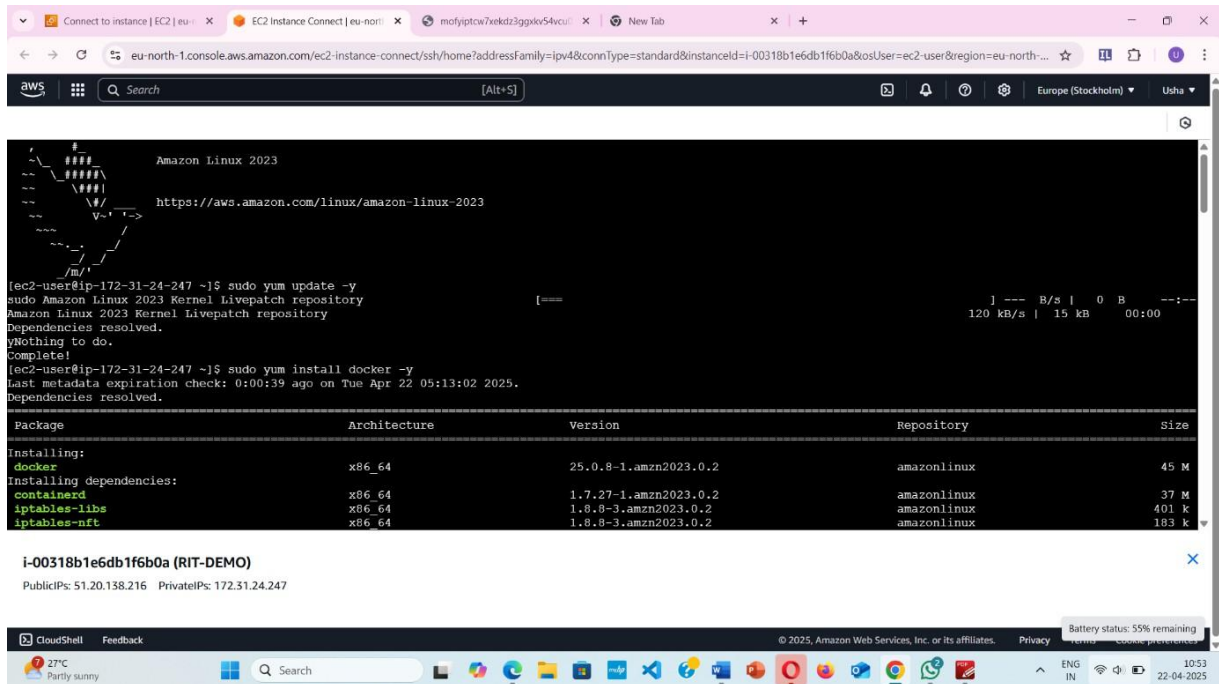
The screenshot shows the AWS Management Console for the eu-north-1 region. The 'Instances' page displays a list of EC2 instances. The instance 'i-00318b1e6db1f6b0a' (RIT-DEMO) is selected. The instance details show it is in the 'Running' state, using the 't3.micro' instance type, and has a public IPv4 address of 51.20.138.216. The instance is located in the eu-north-1 region, availability zone eu-north-1a.

Press connect



The screenshot shows the 'Connect to instance' page in the AWS Management Console. The instance 'i-00318b1e6db1f6b0a' (RIT-DEMO) is selected. The 'Connect using EC2 Instance Connect' option is chosen. The 'Public IPv4 address' is 51.20.138.216. The 'Username' is 'ec2-user'.

It will take you to Linux Machine



```

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-24-247 ~]$ sudo yum update -y
sudo Amazon Linux 2023 Kernel Livepatch repository
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-24-247 ~]$ sudo yum install docker -y
last metadata expiration check: 0:00:39 ago on Tue Apr 22 05:13:02 2025.
Dependencies resolved.

Package      Architecture Version      Repository      Size
Installing:
docker        x86_64      25.0.8-1.amzn2023.0.2  amazonlinux      45 M
Installing dependencies:
containerd    x86_64      1.7.27-1.amzn2023.0.2  amazonlinux      37 M
iptables-libs x86_64      1.8.0-3.amzn2023.0.2  amazonlinux      401 k
iptables-nft  x86_64      1.8.0-3.amzn2023.0.2  amazonlinux      183 k

i-00318b1e6db1f6b0a (RIT-DEMO)
PublicIPs: 51.20.138.216 PrivateIPs: 172.31.24.247
  
```

EXECUTE BELOW COMMANDS IN LINUX ENGINE

sudo yum update -y

sudo yum install docker -y

sudo service docker start

sudo service docker status

(ctrl Z)

sudo su

docker version

docker pull nginx

docker images

docker run -d -p 80:80 nginx

docker ps

after this go to ec2 instance → connect → copy public ip



CLOUD COMPUTING LABORATORY (BCS601)

Connect to instance **Info**

Connect to your instance i-00318b1e6db1f6b0a (RIT-DEMO) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-00318b1e6db1f6b0a (RIT-DEMO)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

☒ **Public IPv4 address**
51.20.138.216

☐ **IPv6 address**
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

ec2-user

Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

Paste it in browser

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

